

The Notebooks • Era V — Prove

Cream construction folio

Testing Mode • UK • IB • US

Exercises

T.V.1.001 • GCSE Higher geometric proof write-ups • challenge • UK_GCSE_H

- (a) [never/neutral/2mk] List the three structural steps of mathematical induction.
- (b) [never/UK/3mk] Inductive step only: assume $1 + \dots + k = k(k+1)/2$; prove for $k+1$.
- (c) [never/UK/2mk] Name one write-up error that loses IB AA HL induction marks.

T.V.1.002 • Proof by induction AA HL staple pack • challenge • IB_AA_HL

- (a) [never/neutral/2mk] List the three structural steps of mathematical induction.
- (b) [never/IB/3mk] Inductive step only: assume $1 + \dots + k = k(k+1)/2$; prove for $k+1$.
- (c) [never/IB/2mk] Name one write-up error that loses IB AA HL induction marks.

T.V.1.003 • Induction: inequalities and sequences • challenge • IB_AA_HL

- (a) [never/neutral/2mk] List the three structural steps of mathematical induction.
- (b) [never/IB/3mk] Inductive step only: assume $1 + \dots + k = k(k+1)/2$; prove for $k+1$.
- (c) [never/IB/2mk] Name one write-up error that loses IB AA HL induction marks.

T.V.1.004 • Counterexample hunting exam style • core • UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] List the three structural steps of mathematical induction.
- (b) [never/UK/3mk] Inductive step only: assume $1 + \dots + k = k(k+1)/2$; prove for $k+1$.
- (c) [never/UK/2mk] Name one write-up error that loses IB AA HL induction marks.

T.V.1.005 • Two-column congruence proof drills • core • US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/2mk] List the three structural steps of mathematical induction.
- (b) [never/US/3mk] Inductive step only: assume $1 + \dots + k = k(k+1)/2$; prove for $k+1$.
- (c) [never/US/2mk] Name one write-up error that loses IB AA HL induction marks.

T.V.1.006 • IB exploration mini-write structure • strange • IB_MYP,IB_AA_SL,IB_AA_HL

- (a) [never/neutral/2mk] List the three structural steps of mathematical induction.
- (b) [never/IB/3mk] Inductive step only: assume $1 + \dots + k = k(k+1)/2$; prove for $k+1$.
- (c) [never/IB/2mk] Name one write-up error that loses IB AA HL induction marks.

T.V.1.007 • Indirect proof (contradiction) geometry classic • challenge • US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/2mk] List the three structural steps of mathematical induction.
- (b) [never/US/3mk] Inductive step only: assume $1 + \dots + k = k(k+1)/2$; prove for $k+1$.
- (c) [never/US/2mk] Name one write-up error that loses IB AA HL induction marks.

T.V.2.001 • Parallel line angle chase exam pack • core • UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Parallel lines; corresponding angle to 46° ?
- (b) [never/UK/1mk] Parallel lines; co-interior with 46° ?
- (c) [never/UK/1mk] Find alternate angle to 46° between parallels.

T.V.2.002 • Prove corresponding/alternate equal • core • UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Parallel lines; corresponding angle to 43° ?
- (b) [never/UK/1mk] Parallel lines; co-interior with 43° ?
- (c) [never/UK/1mk] Find alternate angle to 43° between parallels.

T.V.3.001 • Congruence proof multi-step GCSE/US • core • UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Name three congruence conditions for triangles.
- (b) [never/UK/2mk] Does SSA always prove congruence? Why/why not?
- (c) [never/UK/2mk] In «Congruence proof multi-step GCSE/US»: what does CPCTC allow you to conclude after triangles are congruent?

T.V.3.002 · Isosceles triangle theorem proofs · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] [Isosceles triangle theorem proofs] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $3 \times 3 \times 6$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 3) standing on its base.

T.V.3.003 · Points of concurrency justification pack · challenge · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Points of concurrency justification pack»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Points of concurrency justification pack»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Points of concurrency justification pack», then interpret.

T.V.3.004 · Exterior angle of triangle proof applications · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] [Exterior angle of triangle proof applications] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $7 \times 5 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Exterior angle of triangle proof applications»: build the figure, measure, verify the claimed relation.

T.V.3.005 · CPCTC in multi-triangle proofs · challenge · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] [CPCTC in multi-triangle proofs] Which view is the plan — front, side, or top?
- (b) [never/US/1mk] Cuboid $4 \times 2 \times 3$: how many rectangles in a correct net?
- (c) [allowed/US/2mk] Describe the front elevation of a square-based pyramid (base 4) standing on its base.

T.V.4.001 · Interior angle sum proof for polygons · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] [Interior angle sum proof for polygons] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $4 \times 2 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 4) standing on its base.

T.V.4.002 · Regular polygon exterior angle proofs · core · UK_GCSE_H

- (a) [never/neutral/1mk] [Regular polygon exterior angle proofs] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $8 \times 5 \times 6$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 8) standing on its base.

T.V.4.003 · Tessellation and interior angle conditions proof · strange · UK_GCSE_H

- (a) [never/neutral/1mk] [Tessellation and interior angle conditions proof] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $3 \times 2 \times 4$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Tessellation and interior angle conditions proof»: build the figure, measure, verify the claimed relation.

T.V.5.001 · Similar triangles exam multi-step · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Lengths scale by 2. Area scale factor?
- (b) [never/UK/1mk] A side 6 cm becomes ? after enlargement $SF=2$.
- (c) [never/UK/2mk] Volumes scale by ? if lengths scale by 2.

T.V.5.002 · AA/SSS/SAS similarity justification · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «AA/SSS/SAS similarity justification»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «AA/SSS/SAS similarity justification»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AA/SSS/SAS similarity justification», then interpret.

T.V.5.003 · Area ratios of similar shapes exam · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Lengths scale by 2. Area scale factor?
- (b) [never/UK/1mk] A side 6 cm becomes ? after enlargement $SF=2$.
- (c) [never/UK/2mk] Volumes scale by ? if lengths scale by 2.

T.V.5.004 · Intercept theorems (basic proportionality) · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] [Intercept theorems (basic proportionality)] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $5 \times 5 \times 6$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 5) standing on its base.

T.V.5.005 · Shadow and mirror similar-triangle measurement · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Lengths scale by 5. Area scale factor?
- (b) [never/UK/1mk] A side 6 cm becomes ? after enlargement SF=5.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Shadow and mirror similar-triangle measurement»: build the figure, measure, verify the claimed relation.

T.V.5.006 · Parallel lines $\Rightarrow$ similar triangles patterns · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Parallel lines; corresponding angle to 63° ?
- (b) [never/UK/1mk] Parallel lines; co-interior with 63° ?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Parallel lines $\Rightarrow$ similar triangles patterns»: build the figure, measure, verify the claimed relation.

T.V.6.001 · Exact trig values $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$ pack · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.6.002 · Exact value multi-step without calculator · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.6.003 · Angle of elevation multi-step exam · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] From a point 32 m from a tower, angle of elevation to the top is θ . Write $\tan \theta$ in terms of height h .
- (b) [never/UK/2mk] Angle of elevation 30° , distance 32 m. Exact height of tower?
- (c) [allowed/UK/3mk] Observer eye height 1.5 m; angle of elevation to a 32 m tree top from 32 m away. Set up the equation for the angle θ .

T.V.6.004 · Angle of depression with bearings mix · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Write 75° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 075 . Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Angle of depression with bearings mix»: build the figure, measure, verify the claimed relation.

T.V.6.005 · SOHCAHTOA worded GCSE Foundation/Higher · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 45^\circ \cos 45^\circ$.

T.V.6.006 · Find sides with reciprocal ratios · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.6.007 · Trig in isosceles / split base techniques · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 45^\circ \cos 45^\circ$.

T.V.6.008 · 3D trigonometry: angle between line and plane · strange · UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «3D trigonometry: angle between line and plane»: build the figure, measure, verify the claimed relation.

T.V.6.009 · 3D trigonometry: angle between two planes · strange · UK_GCSE_H,IB_AA_HL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «3D trigonometry: angle between two planes»: build the figure, measure, verify the claimed relation.

T.V.6.010 · 3D Pythagoras + trig combined · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Legs 7 cm, 24 cm. Hypotenuse?
- (b) [never/UK/2mk] Hypotenuse 25 m, leg 7 m. Other leg?
- (c) [allowed/UK/2mk] Ladder 25 m; base 7 m from wall. Reach height?

T.V.6.011 · Right-trig calculator vs exact decision · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.6.012 · Inverse trig range awareness (principal values) · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.6.013 · Clinometer / practical elevation contexts · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] From a point 27 m from a tower, angle of elevation to the top is θ . Write $\tan \theta$ in terms of height h .
- (b) [never/UK/2mk] Angle of elevation 30° , distance 27 m. Exact height of tower?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Clinometer / practical elevation contexts»: build the figure, measure, verify the claimed relation.

T.V.6.014 · Trig ratios as slopes and unit rates · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/US/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/US/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.6.015 · Exact tan 30/45/60 without calculator · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.6.016 · Two-triangle elevation problems · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] From a point 21 m from a tower, angle of elevation to the top is θ . Write $\tan \theta$ in terms of height h .
- (b) [never/UK/2mk] Angle of elevation 30° , distance 21 m. Exact height of tower?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Two-triangle elevation problems»: build the figure, measure, verify the claimed relation.

T.V.6.017 · Bearing and distance from trig in right Δ · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Write 200° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 200° . Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearing and distance from trig in right Δ »: build the figure, measure, verify the claimed relation.

T.V.6.018 · Trigonometric ratios of complementary angles · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.6.019 · Exact values for related angles $120^\circ, 135^\circ, 150^\circ$ · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Unit-circle lab for «Exact values for related angles $120^\circ, 135^\circ, 150^\circ$ »: sample three angles; record ($\cos, \sin$).

T.V.6.020 · Slope angle of a roof / ramp accessibility · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.6.021 · Trigonometry non-calc paper strategies · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.6.022 · Double triangle ladder against wall classic · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Double triangle ladder against wall classic»: build the figure, measure, verify the claimed relation.

T.V.6.023 · Trigonometry in non-right by dropping altitude · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Trigonometry in non-right by dropping altitude»: build the figure, measure, verify the claimed relation.

T.V.6.024 · Inclination of a line from $\tan \theta = m$ · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 45^\circ \cos 45^\circ$.

T.V.6.025 · Exact values application set 1 · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.6.026 · Exact values application set 2 · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.6.027 · Exact values application set 3 · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.6.028 · Exact values application set 4 · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.6.029 · 3D trig optional challenge 1 · strange · UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «3D trig optional challenge 1»: build the figure, measure, verify the claimed relation.

T.V.6.030 · 3D trig optional challenge 2 · strange · UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «3D trig optional challenge 2»: build the figure, measure, verify the claimed relation.

T.V.6.031 · 3D trig optional challenge 3 · strange · UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «3D trig optional challenge 3»: build the figure, measure, verify the claimed relation.

T.V.7.001 · Bearings: three-figure notation drills · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Write 145° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 145. Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearings: three-figure notation drills»: build the figure, measure, verify the claimed relation.

T.V.7.002 · Bearings: reverse bearings (+180° rule) · core · UK_GCSE_H

- (a) [never/neutral/1mk] Write 35° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 035. Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearings: reverse bearings (+180° rule)»: build the figure, measure, verify the claimed relation.

T.V.7.003 · Bearings: multi-leg route problems · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Write 310° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 310. Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearings: multi-leg route problems»: build the figure, measure, verify the claimed relation.

T.V.7.004 · Bearings: exam worded navigation Qs · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Write 145° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 145. Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearings: exam worded navigation Qs»: build the figure, measure, verify the claimed relation.

T.V.7.005 · Bearings with cosine/sine rule · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Write 35° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 035. Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearings with cosine/sine rule»: build the figure, measure, verify the claimed relation.

T.V.7.006 · Law of sines ambiguous case SSA deep pack · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Write 255° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 255. Bearing of A from B?
- (c) [allowed/UK/2mk] Ship sails 255 for 12 km then 345 for 5 km. Name the rule to find straight-line return distance.

T.V.7.007 · Law of cosines rearrange for angles · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Write 310° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 310. Bearing of A from B?
- (c) [allowed/UK/2mk] Ship sails 310 for 12 km then 040 for 5 km. Name the rule to find straight-line return distance.

T.V.7.008 · Area $\frac{1}{2}ab \sin C$ exam pack · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Write 255° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 255. Bearing of A from B?
- (c) [allowed/UK/2mk] Ship sails 255 for 12 km then 345 for 5 km. Name the rule to find straight-line return distance.

T.V.7.009 · Heron vs sine-area method choice · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Write 255° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 255. Bearing of A from B?
- (c) [allowed/UK/2mk] Ship sails 255 for 12 km then 345 for 5 km. Name the rule to find straight-line return distance.

T.V.7.010 · Surveying triangulation exam contexts · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Write 48° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 048. Bearing of A from B?
- (c) [allowed/UK/2mk] Ship sails 048 for 12 km then 138 for 5 km. Name the rule to find straight-line return distance.

T.V.7.011 · Solve any triangle mixed timed set · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Write 110° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 110. Bearing of A from B?
- (c) [allowed/UK/2mk] Ship sails 110 for 12 km then 200 for 5 km. Name the rule to find straight-line return distance.

T.V.7.012 · Area of triangle given SAS multi-step · core · UK_GCSE_H

- (a) [never/neutral/1mk] Write 48° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 048. Bearing of A from B?
- (c) [allowed/UK/2mk] Ship sails 048 for 12 km then 138 for 5 km. Name the rule to find straight-line return distance.

T.V.7.013 · Obtuse triangles cosine rule careful signs · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Write 255° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 255. Bearing of A from B?
- (c) [allowed/UK/2mk] Ship sails 255 for 12 km then 345 for 5 km. Name the rule to find straight-line return distance.

T.V.7.014 · Navigation: wind/current vector triangle · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] $a=(6, -1)$, $b=(5, 5)$. Find $a+b$ as a column vector.
- (b) [never/UK/2mk] $|a|$ for $a=(6, -1)$?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Navigation: wind/current vector triangle»: build the figure, measure, verify the claimed relation.

T.V.7.015 · Ambiguous case: height vs given side decision tree · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Write 62° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 062. Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Ambiguous case: height vs given side decision tree»: build the figure, measure, verify the claimed relation.

T.V.7.016 · Largest angle opposite largest side justification · core · UK_GCSE_H

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Largest angle opposite largest side justification»?
- (b) [allowed/UK/2mk] Write a justification sentence frame for «Largest angle opposite largest side justification»: Because ____, by ____, therefore ____.
- (c) [allowed/UK/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Largest angle opposite largest side justification», then interpret.

T.V.7.017 · Flight path bearings with ground speed · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Write 110° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 110. Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Flight path bearings with ground speed»: build the figure, measure, verify the claimed relation.

T.V.7.018 · Bearings: airport runway / orienteering contexts · core · UK_GCSE_H

- (a) [never/neutral/1mk] Write 255° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 255. Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearings: airport runway / orienteering contexts»: build the figure, measure, verify the claimed relation.

T.V.7.019 · Bearings navigation paper 1 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Write 75° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 075. Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearings navigation paper 1»: build the figure, measure, verify the claimed relation.

T.V.7.020 · Bearings navigation paper 2 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Write 62° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 062. Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearings navigation paper 2»: build the figure, measure, verify the claimed relation.

T.V.7.021 · Bearings navigation paper 3 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Write 200° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 200. Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearings navigation paper 3»: build the figure, measure, verify the claimed relation.

T.V.7.022 · Bearings navigation paper 4 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Write 310° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 310. Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearings navigation paper 4»: build the figure, measure, verify the claimed relation.

T.V.8.001 · Circle theorems: angle in a semicircle · core · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 40° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [never/UK/2mk] Cyclic quadrilateral; $\angle A = 40^\circ$. Find $\angle C$.

T.V.8.002 · Circle theorems: centre angle is twice · core · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 80° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [never/UK/2mk] Cyclic quadrilateral; $\angle A = 80^\circ$. Find $\angle C$.

T.V.8.003 · Circle theorems: angles in same segment · core · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 72° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [never/UK/2mk] Cyclic quadrilateral; $\angle A = 72^\circ$. Find $\angle C$.

T.V.8.004 · Circle theorems: opposite angles cyclic quad · core · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 80° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [never/UK/2mk] Cyclic quadrilateral; $\angle A = 80^\circ$. Find $\angle C$.

T.V.8.005 · Circle theorems: alternate segment · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 72° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [never/UK/2mk] Cyclic quadrilateral; $\angle A = 72^\circ$. Find $\angle C$.

T.V.8.006 · Circle theorems: tangent-chord angle · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 40° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [never/UK/2mk] Cyclic quadrilateral; $\angle A = 40^\circ$. Find $\angle C$.

T.V.8.007 · Circle theorems mixed proof pack · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 64° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Circle theorems mixed proof pack»: build the figure, measure, verify the claimed relation.

T.V.8.008 · Equation of circle centre origin + tangent · core · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 80° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [never/UK/2mk] Cyclic quadrilateral; $\angle A = 80^\circ$. Find $\angle C$.

T.V.8.009 · Tangent from external point length · core · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 100° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [never/UK/2mk] Cyclic quadrilateral; $\angle A = 100^\circ$. Find $\angle C$.

T.V.8.010 · Intersecting chords length theorem · core · UK_GCSE_H,US_CCSS_HS,UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 40° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [never/UK/2mk] Cyclic quadrilateral; $\angle A = 40^\circ$. Find $\angle C$.

T.V.8.011 · Sector/segment area exact exam · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 64° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [never/UK/2mk] Cyclic quadrilateral; $\angle A = 64^\circ$. Find $\angle C$.

T.V.8.012 · Radians arc/sector IB/AP bridge · core · IB_AA_SL,IB_AA_HL,AP_CALC_AB

- (a) [never/neutral/1mk] Angle at centre 50° . Angle at circumference (same arc)?
- (b) [never/IB/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [never/IB/2mk] Cyclic quadrilateral; $\angle A = 50^\circ$. Find $\angle C$.

T.V.8.013 · Proof: tangent $\perp$ radius · core · UK_GCSE_H,US_CCSS_HS,UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 100° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Proof: tangent $\perp$ radius»: build the figure, measure, verify the claimed relation.

T.V.8.014 · Arc length vs sector vs segment decision · core · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 64° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [never/UK/2mk] Cyclic quadrilateral; $\angle A = 64^\circ$. Find $\angle C$.

T.V.8.015 · Trig in circle: chord length via sine · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Angle at centre 72° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Trig in circle: chord length via sine»: build the figure, measure, verify the claimed relation.

T.V.8.016 · Circle theorems + trig hybrid problems · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 100° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Circle theorems + trig hybrid problems»: build the figure, measure, verify the claimed relation.

T.V.8.017 · Circle theorems: radius to tangent proof write · core · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 64° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Circle theorems: radius to tangent proof write»: build the figure, measure, verify the claimed relation.

T.V.8.018 · Circle theorems mixed figure 1 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 100° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Circle theorems mixed figure 1»: build the figure, measure, verify the claimed relation.

T.V.8.019 · Circle theorems mixed figure 2 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 50° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Circle theorems mixed figure 2»: build the figure, measure, verify the claimed relation.

T.V.8.020 · Circle theorems mixed figure 3 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 50° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Circle theorems mixed figure 3»: build the figure, measure, verify the claimed relation.

T.V.8.021 · Circle theorems mixed figure 4 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 80° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Circle theorems mixed figure 4»: build the figure, measure, verify the claimed relation.

T.V.8.022 · Circle theorems mixed figure 5 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Angle at centre 80° . Angle at circumference (same arc)?
- (b) [never/UK/2mk] Triangle in a semicircle; one acute angle 32° . Other acute angle?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Circle theorems mixed figure 5»: build the figure, measure, verify the claimed relation.

T.V.9.001 · Unit circle: verticality of sine · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Unit circle: coordinates at 135° ?
- (b) [never/UK/1mk] Reference angle for 135° ?
- (c) [allowed/UK/3mk ■ LAB] Unit-circle lab for «Unit circle: verticality of sine»: sample three angles; record (cos,sin).

T.V.9.002 · Unit circle: horizontality of cosine · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Unit circle: coordinates at 0° ?
- (b) [never/UK/1mk] Reference angle for 0° ?
- (c) [allowed/UK/3mk ■ LAB] Unit-circle lab for «Unit circle: horizontality of cosine»: sample three angles; record (cos,sin).

T.V.9.003 · Unit circle: tangent as slope of radius · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 135° ?
- (b) [never/IB/1mk] Reference angle for 135° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle: tangent as slope of radius»: sample three angles; record (cos,sin).

T.V.9.004 · Unit circle special angles exact coordinates · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Unit circle: coordinates at 120° ?
- (b) [never/UK/1mk] Reference angle for 120° ?
- (c) [allowed/UK/3mk ■ LAB] Unit-circle lab for «Unit circle special angles exact coordinates»: sample three angles; record (cos,sin).

T.V.9.005 · Reference angles across four quadrants drills · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Unit circle: coordinates at 150° ?
- (b) [never/UK/1mk] Reference angle for 150° ?
- (c) [allowed/UK/3mk ■ LAB] Unit-circle lab for «Reference angles across four quadrants drills»: sample three angles; record (cos,sin).

T.V.9.006 · ASTC / CAST diagram fluency (UK) · core · UK_GCSE_H

- (a) [never/neutral/1mk] Unit circle: coordinates at 60° ?
- (b) [never/UK/1mk] Reference angle for 60° ?
- (c) [allowed/UK/3mk ■ LAB] Unit-circle lab for «ASTC / CAST diagram fluency (UK)»: sample three angles; record (cos,sin).

T.V.9.007 · Coterminal angles in degrees and radians · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 60° ?
- (b) [never/IB/1mk] Reference angle for 60° ?
- (c) [never/IB/2mk] Using CAST/ASTC, sign of $\cos 60^\circ$?

T.V.9.008 · Radians as arc length on unit circle · core · IB_AA_SL,IB_AA_HL,AP_CALC_AB

- (a) [never/neutral/1mk] Unit circle: coordinates at 135° ?
- (b) [never/IB/1mk] Reference angle for 135° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Radians as arc length on unit circle»: sample three angles; record (cos,sin).

T.V.9.009 · Graph sine: build from unit circle animation · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 0° ?
- (b) [never/IB/1mk] Reference angle for 0° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Graph sine: build from unit circle animation»: sample three angles; record (cos,sin).

T.V.9.010 · Graph cosine: build from unit circle animation · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

(a) [never/neutral/1mk] Unit circle: coordinates at 120° ?

(b) [never/IB/1mk] Reference angle for 120° ?

(c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Graph cosine: build from unit circle animation»: sample three angles; record (cos,sin).

T.V.9.011 · Amplitude/period/phase exam transformations · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Unit circle: coordinates at 120° ?

(b) [never/UK/1mk] Reference angle for 120° ?

(c) [allowed/UK/3mk ■ LAB] Desmos lab for «Amplitude/period/phase exam transformations»: vary one parameter, capture before/after, write one observation.

T.V.9.012 · Desmos: match equation to trig graph challenges · challenge · IB_AA_SL,IB_AA_HL,US_CCSS_HS

(a) [never/neutral/1mk] Unit circle: coordinates at 135° ?

(b) [never/IB/1mk] Reference angle for 135° ?

(c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos: match equation to trig graph challenges»: vary one parameter, capture before/after, write one observation.

T.V.9.013 · Periodic modelling: tides and ferris wheels · core · IB_AI_SL,IB_AI_HL,AP_CALC_AB

(a) [never/neutral/1mk] Unit circle: coordinates at 60° ?

(b) [never/IB/1mk] Reference angle for 60° ?

(c) [allowed/IB/3mk ■ LAB] Desmos lab for «Periodic modelling: tides and ferris wheels»: vary one parameter, capture before/after, write one observation.

T.V.9.014 · Graph tangent with asymptotes exam · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Unit circle: coordinates at 60° ?

(b) [never/UK/1mk] Reference angle for 60° ?

(c) [allowed/UK/3mk ■ LAB] Desmos lab for «Graph tangent with asymptotes exam»: vary one parameter, capture before/after, write one observation.

T.V.9.015 · Inverse trig graphs and ranges · core · IB_AA_SL,IB_AA_HL,AP_CALC_AB

(a) [never/neutral/1mk] Unit circle: coordinates at 60° ?

(b) [never/IB/1mk] Reference angle for 60° ?

(c) [allowed/IB/3mk ■ LAB] Desmos lab for «Inverse trig graphs and ranges»: vary one parameter, capture before/after, write one observation.

T.V.9.016 · Composition arcsin(sin θ) pitfalls · challenge · IB_AA_SL,IB_AA_HL,AP_CALC_AB

(a) [never/neutral/1mk] Unit circle: coordinates at 150° ?

(b) [never/IB/1mk] Reference angle for 150° ?

(c) [never/IB/2mk] Using CAST/ASTC, sign of $\cos 150^\circ$?

T.V.9.017 · Desmos: explore a sin(bx+c)+d parameters live · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

(a) [never/neutral/1mk] Unit circle: coordinates at 30° ?

(b) [never/IB/1mk] Reference angle for 30° ?

(c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos: explore a sin(bx+c)+d parameters live»: vary one parameter, capture before/after, write one observation.

T.V.9.018 · Unit circle: project to axes interactively · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

(a) [never/neutral/1mk] Unit circle: coordinates at 90° ?

(b) [never/IB/1mk] Reference angle for 90° ?

(c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle: project to axes interactively»: sample three angles; record (cos,sin).

T.V.9.019 · Negative angles on unit circle · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

(a) [never/neutral/1mk] Unit circle: coordinates at 0° ?

(b) [never/IB/1mk] Reference angle for 0° ?

(c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Negative angles on unit circle»: sample three angles; record (cos,sin).

T.V.9.020 · Co-terminal and reference mixed mega-drill · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Unit circle: coordinates at 180° ?

(b) [never/UK/1mk] Reference angle for 180° ?

(c) [allowed/UK/3mk ■ LAB] Unit-circle lab for «Co-terminal and reference mixed mega-drill»: sample three angles; record (cos,sin).

T.V.9.021 · Desmos: compare $\sin x$ vs $\sin(2x)$ vs $2\sin x$ · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

(a) [never/neutral/1mk] Unit circle: coordinates at 0° ?

(b) [never/IB/1mk] Reference angle for 0° ?

(c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos: compare $\sin x$ vs $\sin(2x)$ vs $2\sin x$ »: vary one parameter, capture before/after, write one observation.

T.V.9.022 · Unit circle: quiz coordinates at $\pi/6$ multiples · core · IB_AA_SL,IB_AA_HL,AP_CALC_AB

(a) [never/neutral/1mk] Unit circle: coordinates at 180° ?

(b) [never/IB/1mk] Reference angle for 180° ?

(c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle: quiz coordinates at $\pi/6$ multiples»: sample three angles; record (cos,sin).

T.V.9.023 · Convert DMS $\leftrightarrow$ decimal degrees exam · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Unit circle: coordinates at 30° ?

(b) [never/UK/1mk] Reference angle for 30° ?

(c) [never/UK/2mk] Using CAST/ASTC, sign of $\cos 30^\circ$?

T.V.9.024 · Angular speed and linear speed link · challenge · IB_AI_SL,IB_AI_HL,AP_CALC_AB

(a) [never/neutral/1mk] Unit circle: coordinates at 60° ?

(b) [never/IB/1mk] Reference angle for 60° ?

(c) [never/IB/2mk] Using CAST/ASTC, sign of $\cos 60^\circ$?

T.V.9.025 · Graph cosec/sec/cot from sine/cosine · challenge · IB_AA_SL,IB_AA_HL,US_CCSS_HS

(a) [never/neutral/1mk] Unit circle: coordinates at 120° ?

(b) [never/IB/1mk] Reference angle for 120° ?

(c) [allowed/IB/3mk ■ LAB] Desmos lab for «Graph cosec/sec/cot from sine/cosine»: vary one parameter, capture before/after, write one observation.

T.V.9.026 · Phase shift left/right common student errors · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

(a) [never/neutral/1mk] Unit circle: coordinates at 120° ?

(b) [never/US/1mk] Reference angle for 120° ?

(c) [allowed/US/3mk ■ LAB] Desmos lab for «Phase shift left/right common student errors»: vary one parameter, capture before/after, write one observation.

T.V.9.027 · Interactive: drag angle see sin/cos update · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

(a) [never/neutral/1mk] Unit circle: coordinates at 135° ?

(b) [never/IB/1mk] Reference angle for 135° ?

(c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Interactive: drag angle see sin/cos update»: sample three angles; record (cos,sin).

T.V.9.028 · Fourier intuition: sum of sines preview · strange · IB_AA_HL,AP_CALC_BC,IB_AA_HL

(a) [never/neutral/1mk] Unit circle: coordinates at 180° ?

(b) [never/IB/1mk] Reference angle for 180° ?

(c) [allowed/IB/3mk ■ LAB] Desmos lab for «Fourier intuition: sum of sines preview»: vary one parameter, capture before/after, write one observation.

T.V.9.029 · Desmos: find period from two zeros · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

(a) [never/neutral/1mk] Unit circle: coordinates at 90° ?

(b) [never/US/1mk] Reference angle for 90° ?

(c) [allowed/US/3mk ■ LAB] Desmos lab for «Desmos: find period from two zeros»: vary one parameter, capture before/after, write one observation.

T.V.9.030 · Unit circle symmetry: odd/even visual proof · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

(a) [never/neutral/1mk] Unit circle: coordinates at 180° ?

(b) [never/IB/1mk] Reference angle for 180° ?

(c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle symmetry: odd/even visual proof»: sample three angles; record (cos,sin).

T.V.9.031 · Model daylight hours with cosine · challenge · IB_AI_SL,IB_AI_HL,AP_CALC_AB

(a) [never/neutral/1mk] Unit circle: coordinates at 30° ?

(b) [never/IB/1mk] Reference angle for 30° ?

(c) [allowed/IB/3mk ■ LAB] Desmos lab for «Model daylight hours with cosine»: vary one parameter, capture before/after, write one observation.

T.V.9.032 · Radians exclusively: no degree crutches · challenge · IB_AA_SL,IB_AA_HL,AP_CALC_AB

- (a) [never/neutral/1mk] Unit circle: coordinates at 30° ?
- (b) [never/IB/1mk] Reference angle for 30° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Radians exclusively: no degree crutches»: sample three angles; record (cos,sin).

T.V.9.033 · Interactive geometry: isolate sin vs cos projections · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 120° ?
- (b) [never/IB/1mk] Reference angle for 120° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Interactive geometry: isolate sin vs cos projections»: sample three angles; record (cos,sin).

T.V.9.034 · Desmos: solve $\sin x = \cos x$ graphically and exactly · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Unit circle: coordinates at 90° ?
- (b) [never/US/1mk] Reference angle for 90° ?
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Desmos: solve $\sin x = \cos x$ graphically and exactly»: vary one parameter, capture before/after, write one observation.

T.V.9.035 · Amplitude from max–min; midline average · core · IB_AI_SL,IB_AI_HL,AP_CALC_AB

- (a) [never/neutral/1mk] Unit circle: coordinates at 45° ?
- (b) [never/IB/1mk] Reference angle for 45° ?
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Amplitude from max–min; midline average»: vary one parameter, capture before/after, write one observation.

T.V.9.036 · Unit circle: memory palace for special angles · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 150° ?
- (b) [never/IB/1mk] Reference angle for 150° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle: memory palace for special angles»: sample three angles; record (cos,sin).

T.V.9.037 · Desmos: Fourier square-wave tease with odd harmonics · strange · IB_AA_HL,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Unit circle: coordinates at 135° ?
- (b) [never/IB/1mk] Reference angle for 135° ?
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos: Fourier square-wave tease with odd harmonics»: vary one parameter, capture before/after, write one observation.

T.V.9.038 · Unit circle fluency round 1 · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 45° ?
- (b) [never/IB/1mk] Reference angle for 45° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle fluency round 1»: sample three angles; record (cos,sin).

T.V.9.039 · Unit circle fluency round 2 · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 90° ?
- (b) [never/IB/1mk] Reference angle for 90° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle fluency round 2»: sample three angles; record (cos,sin).

T.V.9.040 · Unit circle fluency round 3 · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 150° ?
- (b) [never/IB/1mk] Reference angle for 150° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle fluency round 3»: sample three angles; record (cos,sin).

T.V.9.041 · Unit circle fluency round 4 · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 60° ?
- (b) [never/IB/1mk] Reference angle for 60° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle fluency round 4»: sample three angles; record (cos,sin).

T.V.9.042 · Unit circle fluency round 5 · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 90° ?
- (b) [never/IB/1mk] Reference angle for 90° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle fluency round 5»: sample three angles; record (cos,sin).

T.V.9.043 · Unit circle fluency round 6 · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 60° ?
- (b) [never/IB/1mk] Reference angle for 60° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle fluency round 6»: sample three angles; record (cos,sin).

T.V.9.044 · Unit circle fluency round 7 · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 150° ?
- (b) [never/IB/1mk] Reference angle for 150° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle fluency round 7»: sample three angles; record (cos,sin).

T.V.9.045 · Desmos trig graph challenge 1 · challenge · IB_AA_SL,IB_AA_HL,AP_CALC_AB

- (a) [never/neutral/1mk] Unit circle: coordinates at 180° ?
- (b) [never/IB/1mk] Reference angle for 180° ?
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos trig graph challenge 1»: vary one parameter, capture before/after, write one observation.

T.V.9.046 · Desmos trig graph challenge 2 · challenge · IB_AA_SL,IB_AA_HL,AP_CALC_AB

- (a) [never/neutral/1mk] Unit circle: coordinates at 180° ?
- (b) [never/IB/1mk] Reference angle for 180° ?
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos trig graph challenge 2»: vary one parameter, capture before/after, write one observation.

T.V.9.047 · Desmos trig graph challenge 3 · challenge · IB_AA_SL,IB_AA_HL,AP_CALC_AB

- (a) [never/neutral/1mk] Unit circle: coordinates at 45° ?
- (b) [never/IB/1mk] Reference angle for 45° ?
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos trig graph challenge 3»: vary one parameter, capture before/after, write one observation.

T.V.9.048 · Desmos trig graph challenge 4 · challenge · IB_AA_SL,IB_AA_HL,AP_CALC_AB

- (a) [never/neutral/1mk] Unit circle: coordinates at 45° ?
- (b) [never/IB/1mk] Reference angle for 45° ?
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos trig graph challenge 4»: vary one parameter, capture before/after, write one observation.

T.V.9.049 · Desmos trig graph challenge 5 · challenge · IB_AA_SL,IB_AA_HL,AP_CALC_AB

- (a) [never/neutral/1mk] Unit circle: coordinates at 0° ?
- (b) [never/IB/1mk] Reference angle for 0° ?
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos trig graph challenge 5»: vary one parameter, capture before/after, write one observation.

T.V.9.050 · Desmos trig graph challenge 6 · challenge · IB_AA_SL,IB_AA_HL,AP_CALC_AB

- (a) [never/neutral/1mk] Unit circle: coordinates at 60° ?
- (b) [never/IB/1mk] Reference angle for 60° ?
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos trig graph challenge 6»: vary one parameter, capture before/after, write one observation.

T.V.9.051 · Desmos trig graph challenge 7 · challenge · IB_AA_SL,IB_AA_HL,AP_CALC_AB

- (a) [never/neutral/1mk] Unit circle: coordinates at 120° ?
- (b) [never/IB/1mk] Reference angle for 120° ?
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos trig graph challenge 7»: vary one parameter, capture before/after, write one observation.

T.V.10.001 · Prove identities structured exam write-ups · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Prove identities structured exam write-ups»?
- (b) [allowed/UK/2mk] Write a justification sentence frame for «Prove identities structured exam write-ups»: Because ____, by ____, therefore ____.
- (c) [allowed/UK/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Prove identities structured exam write-ups», then interpret.

T.V.10.002 · Pythagorean identities rearrange toolkit · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Legs 9 cm, 12 cm. Hypotenuse?
- (b) [never/UK/2mk] Hypotenuse 15 m, leg 9 m. Other leg?
- (c) [allowed/UK/2mk] Ladder 15 m; base 9 m from wall. Reach height?

T.V.10.003 · Compound angle applications: expand $\sin(A \pm B)$ · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.10.004 · Compound angle applications: find exact values · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 45^\circ \cos 45^\circ$.

T.V.10.005 · Double-angle in equation solving · core · IB_AA_SL,IB_AA_HL,UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/IB/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/IB/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.10.006 · Half-angle and t-formulae (where on syllabus) · strange · IB_AA_HL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/IB/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/IB/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.10.007 · Harmonic form $R \sin(\theta \pm \alpha)$ exam pack · challenge · UK_GCSE_H,IB_AA_HL

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 45^\circ \cos 45^\circ$.

T.V.10.008 · Graphical solutions of trig equations · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Graphical solutions of trig equations»: vary one parameter, capture before/after, write one observation.

T.V.10.009 · General solutions in degrees and radians · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 45^\circ \cos 45^\circ$.

T.V.10.010 · Trig equations with multiple angles · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.10.011 · Factor formulae applications (HL) · challenge · IB_AA_HL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/IB/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/IB/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.10.012 · Desmos: verify identity numerically then prove · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/IB/1mk] Exact value of $\cos 30^\circ$.
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos: verify identity numerically then prove»: vary one parameter, capture before/after, write one observation.

T.V.10.013 · Exam mixed identity + equation mega-set · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.10.014 · Linear combinations to single sinusoid modelling · core · IB_AI_SL,IB_AI_HL,IB_AA_HL

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/IB/1mk] Exact value of $\cos 45^\circ$.
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Linear combinations to single sinusoid modelling»: vary one parameter, capture before/after, write one observation.

T.V.10.015 · cos(A–B) expansion applications · core · IB_AA_SL,IB_AA_HL,UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/IB/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/IB/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.10.016 · sin(A+B) in wave interference light model · strange · IB_AI_SL,IB_AI_HL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/IB/1mk] Exact value of $\cos 60^\circ$.
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «sin(A+B) in wave interference light model»: vary one parameter, capture before/after, write one observation.

T.V.10.017 · Product-to-sum formulas light use · strange · IB_AA_HL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/IB/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/IB/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.10.018 · Solve trig equations on restricted intervals · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Solve trig equations on restricted intervals»: vary one parameter, capture before/after, write one observation.

T.V.10.019 · Identity proof: split into cases strategy · challenge · IB_AA_SL,IB_AA_HL,UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/IB/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/IB/2mk] Without calculator, evaluate $2 \sin 45^\circ \cos 45^\circ$.

T.V.10.020 · Equations: quadratic in sin/cos · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 45^\circ \cos 45^\circ$.

T.V.10.021 · Compound angle: expand cos(A+B) then simplify · core · IB_AA_SL,IB_AA_HL,UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/IB/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/IB/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.10.022 · Prove sin(A+B) from area or Ptolemy light · strange · IB_AA_HL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/IB/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/IB/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.10.023 · Exam: which identity first? decision drills · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.10.024 · Hard identity: multi-formula chain · challenge · IB_AA_SL,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/IB/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/IB/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.10.025 · Identities exam pack 1 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.10.026 · Identities exam pack 2 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.10.027 · Identities exam pack 3 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 60^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.10.028 · Identities exam pack 4 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.10.029 · Identities exam pack 5 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 30^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.10.030 · Trig equations graphical+algebraic 1 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 45^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Trig equations graphical+algebraic 1»: vary one parameter, capture before/after, write one observation.

T.V.10.031 · Trig equations graphical+algebraic 2 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Trig equations graphical+algebraic 2»: vary one parameter, capture before/after, write one observation.

T.V.10.032 · Trig equations graphical+algebraic 3 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Trig equations graphical+algebraic 3»: vary one parameter, capture before/after, write one observation.

T.V.10.033 · Trig equations graphical+algebraic 4 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 30^\circ$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Trig equations graphical+algebraic 4»: vary one parameter, capture before/after, write one observation.

T.V.11.001 · Coordinate proofs of parallelograms/rhombus · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [allowed/neutral/1mk] Notation/definition check for «Coordinate proofs of parallelograms/rhombus»: state the key idea in ≤ 20 words.
- (b) [allowed/US/2mk] Standard exam-style: Apply «Coordinate proofs of parallelograms/rhombus» in a worked example that uses the numbers 22 and 9. Show steps.
- (c) [allowed/US/3mk] US dialect challenge: Write and solve one markable question targeting «Coordinate proofs of parallelograms/rhombus» (include answer).

T.V.11.002 · Equation of perpendicular bisector loci link · core · UK_GCSE_H

- (a) [allowed/neutral/1mk] Notation/definition check for «Equation of perpendicular bisector loci link»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Equation of perpendicular bisector loci link» in a worked example that uses the numbers 12 and 7. Show steps.
- (c) [allowed/UK/3mk] UK dialect challenge: Write and solve one markable question targeting «Equation of perpendicular bisector loci link» (include answer).

T.V.11.003 · G-GPE: circle and parabola algebraically · core · US_CCSS_HS

- (a) [allowed/neutral/1mk] Notation/definition check for «G-GPE: circle and parabola algebraically»: state the key idea in ≤ 20 words.
- (b) [allowed/US/2mk] Standard exam-style: Apply «G-GPE: circle and parabola algebraically» in a worked example that uses the numbers 20 and 16. Show steps.
- (c) [allowed/US/3mk] US dialect challenge: Write and solve one markable question targeting «G-GPE: circle and parabola algebraically» (include answer).

T.V.11.004 · Partition section formula internal/external · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Notation/definition check for «Partition section formula internal/external»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Partition section formula internal/external» in a worked example that uses the numbers 13 and 12. Show steps.
- (c) [allowed/UK/3mk] UK dialect challenge: Write and solve one markable question targeting «Partition section formula internal/external» (include answer).

T.V.12.001 · Complete square to identify conics · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [allowed/neutral/1mk] Notation/definition check for «Complete square to identify conics»: state the key idea in ≤ 20 words.
- (b) [allowed/US/2mk] Standard exam-style: Apply «Complete square to identify conics» in a worked example that uses the numbers 17 and 5. Show steps.
- (c) [allowed/US/3mk] US dialect challenge: Write and solve one markable question targeting «Complete square to identify conics» (include answer).

T.V.12.002 · Eccentricity and focus-directrix exam (light) · strange · IB_AA_HL,US_CCSS_HS

- (a) [allowed/neutral/1mk] Notation/definition check for «Eccentricity and focus-directrix exam (light)»: state the key idea in ≤ 20 words.
- (b) [allowed/IB/2mk] Standard exam-style: Apply «Eccentricity and focus-directrix exam (light)» in a worked example that uses the numbers 4 and 15. Show steps.
- (c) [allowed/IB/3mk] IB dialect challenge: Write and solve one markable question targeting «Eccentricity and focus-directrix exam (light)» (include answer).

T.V.13.001 · Describe fully: translation reflection rotation enlargement · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Notation/definition check for «Describe fully: translation reflection rotation enlargement»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Describe fully: translation reflection rotation enlargement» in a worked example that uses the numbers 22 and 14. Show steps.
- (c) [allowed/UK/3mk] UK dialect challenge: Write and solve one markable question targeting «Describe fully: translation reflection rotation enlargement» (include answer).

T.V.13.002 · Invariant points and lines under transforms · challenge · UK_GCSE_H

- (a) [allowed/neutral/1mk] Notation/definition check for «Invariant points and lines under transforms»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Invariant points and lines under transforms» in a worked example that uses the numbers 14 and 15. Show steps.
- (c) [allowed/UK/3mk] UK dialect challenge: Write and solve one markable question targeting «Invariant points and lines under transforms» (include answer).

T.V.13.003 · Matrix representation of 2D transformations · core · IB_AI_HL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix} \times \mathbf{I}$?
- (c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Matrix representation of 2D transformations»: build the figure, measure, verify the claimed relation.

T.V.13.004 · Composition of transformations exam order · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Notation/definition check for «Composition of transformations exam order»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Composition of transformations exam order» in a worked example that uses the numbers 10 and 11. Show steps.
- (c) [allowed/UK/3mk] UK dialect challenge: Write and solve one markable question targeting «Composition of transformations exam order» (include answer).

T.V.13.005 · Enlargement negative SF with centre · challenge · UK_GCSE_H

- (a) [allowed/neutral/1mk] Notation/definition check for «Enlargement negative SF with centre»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Enlargement negative SF with centre» in a worked example that uses the numbers 13 and 5. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Enlargement negative SF with centre»: build the figure, measure, verify the claimed relation.

T.V.14.001 · Plans elevations link to volume estimates · core · UK_GCSE_H

- (a) [never/neutral/1mk] [Plans elevations link to volume estimates] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $4 \times 2 \times 3$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 4) standing on its base.

T.V.14.002 · G-GMD Cavalieri justification language · core · US_CCSS_HS

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «G-GMD Cavalieri justification language»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «G-GMD Cavalieri justification language»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «G-GMD Cavalieri justification language», then interpret.

T.V.14.003 · Density design problems (mass = density×volume) · core · UK_GCSE_H

- (a) [allowed/neutral/1mk] Notation/definition check for «Density design problems (mass = density×volume)»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Density design problems (mass = density×volume)» in a worked example that uses the numbers 20 and 3. Show steps.
- (c) [allowed/UK/3mk] UK dialect challenge: Write and solve one markable question targeting «Density design problems (mass = density×volume)» (include answer).

T.V.14.004 · Frustum volume (where assessed) · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Notation/definition check for «Frustum volume (where assessed)»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Frustum volume (where assessed)» in a worked example that uses the numbers 14 and 14. Show steps.
- (c) [allowed/UK/3mk] UK dialect challenge: Write and solve one markable question targeting «Frustum volume (where assessed)» (include answer).

T.V.14.005 · Cross-sections of cubes/pyramids exam · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Notation/definition check for «Cross-sections of cubes/pyramids exam»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Cross-sections of cubes/pyramids exam» in a worked example that uses the numbers 18 and 8. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Cross-sections of cubes/pyramids exam»: build the figure, measure, verify the claimed relation.

T.V.15.001 · Exam construction: perpendicular bisector · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Notation/definition check for «Exam construction: perpendicular bisector»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Exam construction: perpendicular bisector» in a worked example that uses the numbers 18 and 16. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Exam construction: perpendicular bisector»: build the figure, measure, verify the claimed relation.

T.V.15.002 · Exam construction: angle bisector · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Notation/definition check for «Exam construction: angle bisector»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Exam construction: angle bisector» in a worked example that uses the numbers 3 and 8. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Exam construction: angle bisector»: build the figure, measure, verify the claimed relation.

T.V.15.003 · Exam construction: equilateral / 60° · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Notation/definition check for «Exam construction: equilateral / 60° »: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Exam construction: equilateral / 60° » in a worked example that uses the numbers 17 and 15. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Exam construction: equilateral / 60° »: build the figure, measure, verify the claimed relation.

T.V.15.004 · Exam construction: regular hexagon in circle · core · UK_GCSE_H

- (a) [allowed/neutral/1mk] Notation/definition check for «Exam construction: regular hexagon in circle»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Exam construction: regular hexagon in circle» in a worked example that uses the numbers 17 and 11. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Exam construction: regular hexagon in circle»: build the figure, measure, verify the claimed relation.

T.V.15.005 · Exam construction: parallel through point · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Notation/definition check for «Exam construction: parallel through point»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Exam construction: parallel through point» in a worked example that uses the numbers 21 and 11. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Exam construction: parallel through point»: build the figure, measure, verify the claimed relation.

T.V.15.006 · Loci constructions combined with regions · challenge · UK_GCSE_H

- (a) [allowed/neutral/1mk] Notation/definition check for «Loci constructions combined with regions»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Loci constructions combined with regions» in a worked example that uses the numbers 11 and 3. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Loci constructions combined with regions»: build the figure, measure, verify the claimed relation.

T.V.15.007 · Construct 45° and 90° exam techniques · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Notation/definition check for «Construct 45° and 90° exam techniques»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Construct 45° and 90° exam techniques» in a worked example that uses the numbers 8 and 4. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Construct 45° and 90° exam techniques»: build the figure, measure, verify the claimed relation.

T.V.15.008 · Construct triangle given ASA accurately · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Notation/definition check for «Construct triangle given ASA accurately»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Construct triangle given ASA accurately» in a worked example that uses the numbers 20 and 5. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Construct triangle given ASA accurately»: build the figure, measure, verify the claimed relation.

T.V.15.009 · Construction accuracy drill 1 · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Notation/definition check for «Construction accuracy drill 1»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Construction accuracy drill 1» in a worked example that uses the numbers 7 and 14. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Construction accuracy drill 1»: build the figure, measure, verify the claimed relation.

T.V.15.010 · Construction accuracy drill 2 · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Notation/definition check for «Construction accuracy drill 2»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Construction accuracy drill 2» in a worked example that uses the numbers 10 and 6. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Construction accuracy drill 2»: build the figure, measure, verify the claimed relation.

T.V.15.011 · Construction accuracy drill 3 · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [allowed/neutral/1mk] Notation/definition check for «Construction accuracy drill 3»: state the key idea in ≤ 20 words.
- (b) [allowed/UK/2mk] Standard exam-style: Apply «Construction accuracy drill 3» in a worked example that uses the numbers 6 and 10. Show steps.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Construction accuracy drill 3»: build the figure, measure, verify the claimed relation.

T.V.9.052 · Unit circle: verticality drill with readout · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 120° ?
- (b) [never/IB/1mk] Reference angle for 120° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle: verticality drill with readout»: sample three angles; record (cos,sin).

T.V.9.053 · Unit circle: horizontality drill with readout · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Unit circle: coordinates at 150° ?
- (b) [never/IB/1mk] Reference angle for 150° ?
- (c) [allowed/IB/3mk ■ LAB] Unit-circle lab for «Unit circle: horizontality drill with readout»: sample three angles; record (cos,sin).

T.V.9.054 · Desmos: rebuild tide model from data points · challenge · IB_AI_SL,IB_AI_HL,AP_CALC_AB

- (a) [never/neutral/1mk] Unit circle: coordinates at 180° ?
- (b) [never/IB/1mk] Reference angle for 180° ?
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Desmos: rebuild tide model from data points»: vary one parameter, capture before/after, write one observation.

T.V.10.034 · Compound angle application: $\sin(75^\circ)$ exact · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 60^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 60^\circ \cos 60^\circ$.

T.V.10.035 · Compound angle application: $\cos(15^\circ)$ exact · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.6.032 · Exact values memory palace exam warm-up · core · UK_GCSE_H

- (a) [never/neutral/1mk] Exact value of $\sin 30^\circ$.
- (b) [never/UK/1mk] Exact value of $\cos 45^\circ$.
- (c) [never/UK/2mk] Without calculator, evaluate $2 \sin 30^\circ \cos 30^\circ$.

T.V.7.023 · Bearings reverse and multi-leg combo challenge · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Write 110° from north as a three-figure bearing.
- (b) [never/UK/2mk] Bearing of B from A is 110° . Bearing of A from B?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Bearings reverse and multi-leg combo challenge»: build the figure, measure, verify the claimed relation.

Answer Key — Era V

Brief solutions. Full reasoning in machine JSON.

T.V.1.001

- (a) Base case; inductive hypothesis; inductive step (+ conclusion) — *Standard structure.*
- (b) $LHS=k(k+1)/2+(k+1)=(k+1)(k+2)/2$ — *Add $k+1$; factor.*
- (c) e.g. missing base case / assuming the result / no conclusion — *Structure marks.*

T.V.1.002

- (a) Base case; inductive hypothesis; inductive step (+ conclusion) — *Standard structure.*
- (b) $LHS=k(k+1)/2+(k+1)=(k+1)(k+2)/2$ — *Add $k+1$; factor.*
- (c) e.g. missing base case / assuming the result / no conclusion — *Structure marks.*

T.V.1.003

- (a) Base case; inductive hypothesis; inductive step (+ conclusion) — *Standard structure.*
- (b) $LHS=k(k+1)/2+(k+1)=(k+1)(k+2)/2$ — *Add $k+1$; factor.*
- (c) e.g. missing base case / assuming the result / no conclusion — *Structure marks.*

T.V.1.004

- (a) Base case; inductive hypothesis; inductive step (+ conclusion) — *Standard structure.*
- (b) $LHS=k(k+1)/2+(k+1)=(k+1)(k+2)/2$ — *Add $k+1$; factor.*
- (c) e.g. missing base case / assuming the result / no conclusion — *Structure marks.*

T.V.1.005

- (a) Base case; inductive hypothesis; inductive step (+ conclusion) — *Standard structure.*
- (b) $LHS=k(k+1)/2+(k+1)=(k+1)(k+2)/2$ — *Add $k+1$; factor.*
- (c) e.g. missing base case / assuming the result / no conclusion — *Structure marks.*

T.V.1.006

- (a) Base case; inductive hypothesis; inductive step (+ conclusion) — *Standard structure.*
- (b) $LHS=k(k+1)/2+(k+1)=(k+1)(k+2)/2$ — *Add $k+1$; factor.*
- (c) e.g. missing base case / assuming the result / no conclusion — *Structure marks.*

T.V.1.007

- (a) Base case; inductive hypothesis; inductive step (+ conclusion) — *Standard structure.*
- (b) $LHS=k(k+1)/2+(k+1)=(k+1)(k+2)/2$ — *Add $k+1$; factor.*
- (c) e.g. missing base case / assuming the result / no conclusion — *Structure marks.*

T.V.2.001

- (a) 46° — *Corresponding equal.*
- (b) 134° — *Sum 180° .*
- (c) 46° — *Alternate equal.*

T.V.2.002

- (a) 43° — *Corresponding equal.*
- (b) 137° — *Sum 180° .*
- (c) 43° — *Alternate equal.*

T.V.3.001

- (a) SSS, SAS, ASA (and RHS) — *Standard conditions.*
- (b) No — ambiguous case possible — *SSA not a congruence condition.*
- (c) Corresponding parts of congruent triangles are congruent — *CPCTC.*

T.V.3.002

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.V.3.003

- (a) The governing definition/theorem for Points of concurrency justification pack — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.V.3.004

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.3.005

- (a) Top view (from above) — *Plan = bird's-eye.*

- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.V.4.001

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.V.4.002

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.V.4.003

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.5.001

- (a) 4 — SF^2 .
- (b) 12 cm — $\times SF$.
- (c) 8 — SF^3 .

T.V.5.002

- (a) The governing definition/theorem for AA/SSS/SAS similarity justification — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.V.5.003

- (a) 4 — SF^2 .
- (b) 12 cm — $\times SF$.
- (c) 8 — SF^3 .

T.V.5.004

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.V.5.005

- (a) 25 — SF^2 .
- (b) 30 cm — $\times SF$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.5.006

- (a) 63° — *Corresponding equal.*
- (b) 117° — *Sum 180° .*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.6.001

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.002

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.003

- (a) $\tan \theta = h/d$ — *Opposite/adjacent.*
- (b) $32/\sqrt{3}$ m or $32\sqrt{3}/3$ m — *$\tan 30 = 1/\sqrt{3} = h/d$.*
- (c) $\tan \theta = (32-1.5)/32$ — *Adjusted opposite side.*

T.V.6.004

- (a) 075 — *Pad to 3 digits; clockwise from north.*
- (b) 255 — $\pm 180^\circ$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.6.005

- (a) $\sqrt{2}/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) 1 — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.006

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.007

- (a) $\sqrt{2}/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) 1 — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.008

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.6.009

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.6.010

- (a) 25 cm — $\sqrt{(a^2+b^2)}$.
- (b) 24 m — $\sqrt{(c^2-a^2)}$.
- (c) 24 m — *Pythagoras.*

T.V.6.011

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.012

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.013

- (a) $\tan \theta = h/d$ — *Opposite/adjacent.*
- (b) $27/\sqrt{3}$ m or $27\sqrt{3}/3$ m — $\tan 30 = 1/\sqrt{3} = h/d$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.6.014

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.015

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.016

- (a) $\tan \theta = h/d$ — *Opposite/adjacent.*
- (b) $21/\sqrt{3}$ m or $21\sqrt{3}/3$ m — $\tan 30 = 1/\sqrt{3} = h/d$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.6.017

- (a) 200 — *Pad to 3 digits; clockwise from north.*
- (b) 020 — $\pm 180^\circ$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.6.018

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.019

- (a) $\sqrt{2}/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.6.020

- (a) $1/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.021

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.022

- (a) $1/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.6.023

- (a) $\sqrt{2}/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.6.024

- (a) $\sqrt{2}/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) 1 — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.025

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.026

- (a) $1/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.027

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.028

- (a) $1/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.029

- (a) $\sqrt{2}/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.6.030

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.6.031

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.001

- (a) 145 — *Pad to 3 digits; clockwise from north.*
- (b) 325 — $\pm 180^\circ$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.002

- (a) 035 — *Pad to 3 digits; clockwise from north.*
- (b) 215 — $\pm 180^\circ$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.003

- (a) 310 — *Pad to 3 digits; clockwise from north.*
- (b) 130 — $\pm 180^\circ$.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.004

(a) 145 — *Pad to 3 digits; clockwise from north.*

(b) 325 — $\pm 180^\circ$.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.005

(a) 035 — *Pad to 3 digits; clockwise from north.*

(b) 215 — $\pm 180^\circ$.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.006

(a) 255 — *Pad to 3 digits; clockwise from north.*

(b) 075 — $\pm 180^\circ$.

(c) Cosine rule (with included angle from bearing turn) — *Path triangle; cosine rule.*

T.V.7.007

(a) 310 — *Pad to 3 digits; clockwise from north.*

(b) 130 — $\pm 180^\circ$.

(c) Cosine rule (with included angle from bearing turn) — *Path triangle; cosine rule.*

T.V.7.008

(a) 255 — *Pad to 3 digits; clockwise from north.*

(b) 075 — $\pm 180^\circ$.

(c) Cosine rule (with included angle from bearing turn) — *Path triangle; cosine rule.*

T.V.7.009

(a) 255 — *Pad to 3 digits; clockwise from north.*

(b) 075 — $\pm 180^\circ$.

(c) Cosine rule (with included angle from bearing turn) — *Path triangle; cosine rule.*

T.V.7.010

(a) 048 — *Pad to 3 digits; clockwise from north.*

(b) 228 — $\pm 180^\circ$.

(c) Cosine rule (with included angle from bearing turn) — *Path triangle; cosine rule.*

T.V.7.011

(a) 110 — *Pad to 3 digits; clockwise from north.*

(b) 290 — $\pm 180^\circ$.

(c) Cosine rule (with included angle from bearing turn) — *Path triangle; cosine rule.*

T.V.7.012

(a) 048 — *Pad to 3 digits; clockwise from north.*

(b) 228 — $\pm 180^\circ$.

(c) Cosine rule (with included angle from bearing turn) — *Path triangle; cosine rule.*

T.V.7.013

(a) 255 — *Pad to 3 digits; clockwise from north.*

(b) 075 — $\pm 180^\circ$.

(c) Cosine rule (with included angle from bearing turn) — *Path triangle; cosine rule.*

T.V.7.014

(a) (11, 4) — *Add components.*

(b) $\sqrt{37}$ — $\sqrt{(x^2+y^2)}$.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.015

(a) 062 — *Pad to 3 digits; clockwise from north.*

(b) 242 — $\pm 180^\circ$.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.016

(a) The governing definition/theorem for Largest angle opposite largest side justification — *Name it precisely.*

(b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*

(c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.V.7.017

(a) 110 — *Pad to 3 digits; clockwise from north.*

(b) 290 — $\pm 180^\circ$.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.018

- (a) 255 — *Pad to 3 digits; clockwise from north.*
- (b) 075 — $\pm 180^\circ$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.019

- (a) 075 — *Pad to 3 digits; clockwise from north.*
- (b) 255 — $\pm 180^\circ$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.020

- (a) 062 — *Pad to 3 digits; clockwise from north.*
- (b) 242 — $\pm 180^\circ$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.021

- (a) 200 — *Pad to 3 digits; clockwise from north.*
- (b) 020 — $\pm 180^\circ$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.7.022

- (a) 310 — *Pad to 3 digits; clockwise from north.*
- (b) 130 — $\pm 180^\circ$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.8.001

- (a) 20° — *Centre = $2 \times$ circumference.*
- (b) 58° — *Angle in semicircle 90° ; $180-90-32$.*
- (c) 140° — *Opposite angles sum 180° .*

T.V.8.002

- (a) 40° — *Centre = $2 \times$ circumference.*
- (b) 58° — *Angle in semicircle 90° ; $180-90-32$.*
- (c) 100° — *Opposite angles sum 180° .*

T.V.8.003

- (a) 36° — *Centre = $2 \times$ circumference.*
- (b) 58° — *Angle in semicircle 90° ; $180-90-32$.*
- (c) 108° — *Opposite angles sum 180° .*

T.V.8.004

- (a) 40° — *Centre = $2 \times$ circumference.*
- (b) 58° — *Angle in semicircle 90° ; $180-90-32$.*
- (c) 100° — *Opposite angles sum 180° .*

T.V.8.005

- (a) 36° — *Centre = $2 \times$ circumference.*
- (b) 58° — *Angle in semicircle 90° ; $180-90-32$.*
- (c) 108° — *Opposite angles sum 180° .*

T.V.8.006

- (a) 20° — *Centre = $2 \times$ circumference.*
- (b) 58° — *Angle in semicircle 90° ; $180-90-32$.*
- (c) 140° — *Opposite angles sum 180° .*

T.V.8.007

- (a) 32° — *Centre = $2 \times$ circumference.*
- (b) 58° — *Angle in semicircle 90° ; $180-90-32$.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.8.008

- (a) 40° — *Centre = $2 \times$ circumference.*
- (b) 58° — *Angle in semicircle 90° ; $180-90-32$.*
- (c) 100° — *Opposite angles sum 180° .*

T.V.8.009

- (a) 50° — *Centre = $2 \times$ circumference.*
- (b) 58° — *Angle in semicircle 90° ; $180-90-32$.*
- (c) 80° — *Opposite angles sum 180° .*

T.V.8.010

- (a) 20° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) 140° — Opposite angles sum 180° .

T.V.8.011

- (a) 32° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) 116° — Opposite angles sum 180° .

T.V.8.012

- (a) 25° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) 130° — Opposite angles sum 180° .

T.V.8.013

- (a) 50° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.V.8.014

- (a) 32° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) 116° — Opposite angles sum 180° .

T.V.8.015

- (a) 36° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.V.8.016

- (a) 50° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.V.8.017

- (a) 32° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.V.8.018

- (a) 50° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.V.8.019

- (a) 25° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.V.8.020

- (a) 25° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.V.8.021

- (a) 40° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.V.8.022

- (a) 40° — Centre = $2 \times$ circumference.
- (b) 58° — Angle in semicircle 90° ; $180-90-32$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.V.9.001

- (a) $(-\sqrt{2}/2, \sqrt{2}/2)$ — $(\cos \theta, \sin \theta)$.
- (b) 45° — Acute angle to x-axis.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.V.9.002

- (a) $(1, 0)$ — $(\cos \theta, \sin \theta)$.
- (b) 0° — Acute angle to x-axis.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.003

(a) $(-\sqrt{2}/2, \sqrt{2}/2) — (\cos \theta, \sin \theta).$

(b) $45^\circ — \text{Acute angle to } x\text{-axis}.$

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.004

(a) $(-1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta).$

(b) $60^\circ — \text{Acute angle to } x\text{-axis}.$

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.005

(a) $(\sqrt{3}/2, 1/2) — (\cos \theta, \sin \theta).$

(b) $30^\circ — \text{Acute angle to } x\text{-axis}.$

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.006

(a) $(1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta).$

(b) $60^\circ — \text{Acute angle to } x\text{-axis}.$

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.007

(a) $(1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta).$

(b) $60^\circ — \text{Acute angle to } x\text{-axis}.$

(c) positive — *Quadrant sign chart.*

T.V.9.008

(a) $(-\sqrt{2}/2, \sqrt{2}/2) — (\cos \theta, \sin \theta).$

(b) $45^\circ — \text{Acute angle to } x\text{-axis}.$

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.009

(a) $(1, 0) — (\cos \theta, \sin \theta).$

(b) $0^\circ — \text{Acute angle to } x\text{-axis}.$

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.010

(a) $(-1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta).$

(b) $60^\circ — \text{Acute angle to } x\text{-axis}.$

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.011

(a) $(-1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta).$

(b) $60^\circ — \text{Acute angle to } x\text{-axis}.$

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.012

(a) $(-\sqrt{2}/2, \sqrt{2}/2) — (\cos \theta, \sin \theta).$

(b) $45^\circ — \text{Acute angle to } x\text{-axis}.$

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.013

(a) $(1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta).$

(b) $60^\circ — \text{Acute angle to } x\text{-axis}.$

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.014

(a) $(1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta).$

(b) $60^\circ — \text{Acute angle to } x\text{-axis}.$

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.015

(a) $(1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta).$

(b) $60^\circ — \text{Acute angle to } x\text{-axis}.$

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.016

(a) $(-\sqrt{3}/2, 1/2) — (\cos \theta, \sin \theta).$

(b) $30^\circ — \text{Acute angle to } x\text{-axis}.$

(c) negative — *Quadrant sign chart.*

T.V.9.017

- (a) $(\sqrt{3}/2, 1/2) — (\cos \theta, \sin \theta)$.
 (b) $30^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.018

- (a) $(0, 1) — (\cos \theta, \sin \theta)$.
 (b) $90^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.019

- (a) $(1, 0) — (\cos \theta, \sin \theta)$.
 (b) $0^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.020

- (a) $(-1, 0) — (\cos \theta, \sin \theta)$.
 (b) $0^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.021

- (a) $(1, 0) — (\cos \theta, \sin \theta)$.
 (b) $0^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.022

- (a) $(-1, 0) — (\cos \theta, \sin \theta)$.
 (b) $0^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.023

- (a) $(\sqrt{3}/2, 1/2) — (\cos \theta, \sin \theta)$.
 (b) $30^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) positive — *Quadrant sign chart*.

T.V.9.024

- (a) $(1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta)$.
 (b) $60^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) positive — *Quadrant sign chart*.

T.V.9.025

- (a) $(-1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta)$.
 (b) $60^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.026

- (a) $(-1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta)$.
 (b) $60^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.027

- (a) $(-\sqrt{2}/2, \sqrt{2}/2) — (\cos \theta, \sin \theta)$.
 (b) $45^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.028

- (a) $(-1, 0) — (\cos \theta, \sin \theta)$.
 (b) $0^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.029

- (a) $(0, 1) — (\cos \theta, \sin \theta)$.
 (b) $90^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.030

- (a) $(-1, 0) — (\cos \theta, \sin \theta)$.
 (b) $0^\circ — \text{Acute angle to } x\text{-axis}$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.031

- (a) $(\sqrt{3}/2, 1/2) — (\cos \theta, \sin \theta)$.
- (b) $30^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.032

- (a) $(\sqrt{3}/2, 1/2) — (\cos \theta, \sin \theta)$.
- (b) $30^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.033

- (a) $(-1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta)$.
- (b) $60^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.034

- (a) $(0, 1) — (\cos \theta, \sin \theta)$.
- (b) $90^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.035

- (a) $(\sqrt{2}/2, \sqrt{2}/2) — (\cos \theta, \sin \theta)$.
- (b) $45^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.036

- (a) $(-\sqrt{3}/2, 1/2) — (\cos \theta, \sin \theta)$.
- (b) $30^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.037

- (a) $(-\sqrt{2}/2, \sqrt{2}/2) — (\cos \theta, \sin \theta)$.
- (b) $45^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.038

- (a) $(\sqrt{2}/2, \sqrt{2}/2) — (\cos \theta, \sin \theta)$.
- (b) $45^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.039

- (a) $(0, 1) — (\cos \theta, \sin \theta)$.
- (b) $90^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.040

- (a) $(-\sqrt{3}/2, 1/2) — (\cos \theta, \sin \theta)$.
- (b) $30^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.041

- (a) $(1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta)$.
- (b) $60^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.042

- (a) $(0, 1) — (\cos \theta, \sin \theta)$.
- (b) $90^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.043

- (a) $(1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta)$.
- (b) $60^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.044

- (a) $(-\sqrt{3}/2, 1/2) — (\cos \theta, \sin \theta)$.
- (b) $30^\circ — \text{Acute angle to } x\text{-axis}$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.V.9.045

- (a) $(-1, 0) — (\cos \theta, \sin \theta)$.
- (b) $0^\circ — \text{Acute angle to } x\text{-axis}$.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.046

(a) $(-1, 0)$ — $(\cos \theta, \sin \theta)$.

(b) 0° — *Acute angle to x-axis.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.047

(a) $(\sqrt{2}/2, \sqrt{2}/2)$ — $(\cos \theta, \sin \theta)$.

(b) 45° — *Acute angle to x-axis.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.048

(a) $(\sqrt{2}/2, \sqrt{2}/2)$ — $(\cos \theta, \sin \theta)$.

(b) 45° — *Acute angle to x-axis.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.049

(a) $(1, 0)$ — $(\cos \theta, \sin \theta)$.

(b) 0° — *Acute angle to x-axis.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.050

(a) $(1/2, \sqrt{3}/2)$ — $(\cos \theta, \sin \theta)$.

(b) 60° — *Acute angle to x-axis.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.051

(a) $(-1/2, \sqrt{3}/2)$ — $(\cos \theta, \sin \theta)$.

(b) 60° — *Acute angle to x-axis.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.10.001

(a) The governing definition/theorem for Prove identities structured exam write-ups — *Name it precisely.*

(b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*

(c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.V.10.002

(a) 15 cm — $\sqrt{a^2+b^2}$.

(b) 12 m — $\sqrt{c^2-a^2}$.

(c) 12 m — *Pythagoras.*

T.V.10.003

(a) $1/2$ — *Special triangles.*

(b) $1/2$ — *Exact cosine.*

(c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.004

(a) $\sqrt{2}/2$ — *Special triangles.*

(b) $1/2$ — *Exact cosine.*

(c) 1 — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.005

(a) $\sqrt{3}/2$ — *Special triangles.*

(b) $\sqrt{3}/2$ — *Exact cosine.*

(c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.006

(a) $\sqrt{3}/2$ — *Special triangles.*

(b) $1/2$ — *Exact cosine.*

(c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.007

(a) $\sqrt{2}/2$ — *Special triangles.*

(b) $\sqrt{2}/2$ — *Exact cosine.*

(c) 1 — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.008

(a) $\sqrt{3}/2$ — *Special triangles.*

(b) $\sqrt{2}/2$ — *Exact cosine.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.10.009

- (a) $\sqrt{2}/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) 1 — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.010

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.011

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.012

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.10.013

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.014

- (a) $\sqrt{2}/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.10.015

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.016

- (a) $1/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.10.017

- (a) $1/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.018

- (a) $\sqrt{2}/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.10.019

- (a) $\sqrt{2}/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) 1 — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.020

- (a) $\sqrt{2}/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) 1 — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.021

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.022

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.023

- (a) $1/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.024

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.025

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.026

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.027

- (a) $1/2$ — *Special triangles.*
- (b) $1/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.028

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.029

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.030

- (a) $\sqrt{2}/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.10.031

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.10.032

- (a) $1/2$ — *Special triangles.*
- (b) $\sqrt{2}/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.10.033

- (a) $\sqrt{3}/2$ — *Special triangles.*
- (b) $\sqrt{3}/2$ — *Exact cosine.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.11.001

- (a) Accept accurate paraphrase of: Prove properties using distance/slope/midpoint. — *On-skill definition.*
- (b) Correct worked application of Coordinate proofs of parallelograms/rhombus using 22, 9 — *Use method from «Coordinate proofs of parallelograms/rhombus»; compute carefully with 22, 9.*
- (c) Student-authored on-skill Q + correct solution — *Must use track vocabulary and be markable.*

T.V.11.002

- (a) Accept accurate paraphrase of: Link perp bisector to locus equidistant. — *On-skill definition.*
- (b) Correct worked application of Equation of perpendicular bisector loci link using 12, 7 — *Use method from «Equation of perpendicular bisector loci link»; compute carefully with 12, 7.*
- (c) Student-authored on-skill Q + correct solution — *Must use track vocabulary and be markable.*

T.V.11.003

- (a) Accept accurate paraphrase of: Express geometric properties with equations. — *On-skill definition.*
- (b) Correct worked application of G-GPE: circle and parabola algebraically using 20, 16 — *Use method from «G-GPE: circle and parabola algebraically»; compute carefully with 20, 16.*
- (c) Student-authored on-skill Q + correct solution — *Must use track vocabulary and be markable.*

T.V.11.004

- (a) Accept accurate paraphrase of: Divide a segment in ratio $m:n$. — *On-skill definition.*
- (b) Correct worked application of Partition section formula internal/external using 13, 12 — *Use method from «Partition section formula internal/external»; compute carefully with 13, 12.*
- (c) Student-authored on-skill Q + correct solution — *Must use track vocabulary and be markable.*

T.V.12.001

- (a) Accept accurate paraphrase of: Rewrite general conic; identify type. — *On-skill definition.*
- (b) Correct worked application of Complete square to identify conics using 17, 5 — *Use method from «Complete square to identify conics»; compute carefully with 17, 5.*
- (c) Student-authored on-skill Q + correct solution — *Must use track vocabulary and be markable.*

T.V.12.002

- (a) Accept accurate paraphrase of: Use focus-directrix definition selectively. — *On-skill definition.*
- (b) Correct worked application of Eccentricity and focus-directrix exam (light) using 4, 15 — *Use method from «Eccentricity and focus-directrix exam (light)»; compute carefully with 4, 15.*
- (c) Student-authored on-skill Q + correct solution — *Must use track vocabulary and be markable.*

T.V.13.001

- (a) Accept accurate paraphrase of: GCSE full description mark scheme language. — *On-skill definition.*
- (b) Correct worked application of Describe fully: translation reflection rotation enlargement using 22, 14 — *Use method from «Describe fully: translation reflection rotation enlargement»; compute carefully with 22, 14.*
- (c) Student-authored on-skill Q + correct solution — *Must use track vocabulary and be markable.*

T.V.13.002

- (a) Accept accurate paraphrase of: Find invariants for given transformations. — *On-skill definition.*
- (b) Correct worked application of Invariant points and lines under transforms using 14, 15 — *Use method from «Invariant points and lines under transforms»; compute carefully with 14, 15.*
- (c) Student-authored on-skill Q + correct solution — *Must use track vocabulary and be markable.*

T.V.13.003

- (a) -6 — $ad-bc$.
- (b) $[[1,5],[2,4]]$ — *Identity.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.13.004

- (a) Accept accurate paraphrase of: Compose transforms; order matters. — *On-skill definition.*
- (b) Correct worked application of Composition of transformations exam order using 10, 11 — *Use method from «Composition of transformations exam order»; compute carefully with 10, 11.*
- (c) Student-authored on-skill Q + correct solution — *Must use track vocabulary and be markable.*

T.V.13.005

- (a) Accept accurate paraphrase of: Perform negative enlargements; describe. — *On-skill definition.*
- (b) Correct worked application of Enlargement negative SF with centre using 13, 5 — *Use method from «Enlargement negative SF with centre»; compute carefully with 13, 5.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.14.001

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.V.14.002

- (a) The governing definition/theorem for G-GMD Cavalieri justification language — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.V.14.003

- (a) Accept accurate paraphrase of: GCSE Higher density design contexts. — *On-skill definition.*
- (b) Correct worked application of Density design problems (mass = density $\times$ volume) using 20, 3 — *Use method from «Density design problems (mass = density $\times$ volume)»; compute carefully with 20, 3.*
- (c) Student-authored on-skill Q + correct solution — *Must use track vocabulary and be markable.*

T.V.14.004

- (a) Accept accurate paraphrase of: Find volume of frustum of cone/pyramid. — *On-skill definition.*
- (b) Correct worked application of Frustum volume (where assessed) using 14, 14 — *Use method from «Frustum volume (where assessed)»; compute carefully with 14, 14.*
- (c) Student-authored on-skill Q + correct solution — *Must use track vocabulary and be markable.*

T.V.14.005

- (a) Accept accurate paraphrase of: Predict plane cross-section shapes. — *On-skill definition.*

- (b) Correct worked application of Cross-sections of cubes/pyramids exam using 18, 8 — *Use method from «Cross-sections of cubes/pyramids exam»; compute carefully with 18, 8.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.15.001

- (a) Accept accurate paraphrase of: Construct perp bisector within tolerance. — *On-skill definition.*
- (b) Correct worked application of Exam construction: perpendicular bisector using 18, 16 — *Use method from «Exam construction: perpendicular bisector»; compute carefully with 18, 16.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.15.002

- (a) Accept accurate paraphrase of: Construct angle bisector accurately. — *On-skill definition.*
- (b) Correct worked application of Exam construction: angle bisector using 3, 8 — *Use method from «Exam construction: angle bisector»; compute carefully with 3, 8.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.15.003

- (a) Accept accurate paraphrase of: Construct 60° and equilateral triangle. — *On-skill definition.*
- (b) Correct worked application of Exam construction: equilateral / 60° using 17, 15 — *Use method from «Exam construction: equilateral / 60°»; compute carefully with 17, 15.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.15.004

- (a) Accept accurate paraphrase of: Construct regular hexagon inscribed. — *On-skill definition.*
- (b) Correct worked application of Exam construction: regular hexagon in circle using 17, 11 — *Use method from «Exam construction: regular hexagon in circle»; compute carefully with 17, 11.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.15.005

- (a) Accept accurate paraphrase of: Construct parallel line through a point. — *On-skill definition.*
- (b) Correct worked application of Exam construction: parallel through point using 21, 11 — *Use method from «Exam construction: parallel through point»; compute carefully with 21, 11.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.15.006

- (a) Accept accurate paraphrase of: Construct loci then shade regions. — *On-skill definition.*
- (b) Correct worked application of Loci constructions combined with regions using 11, 3 — *Use method from «Loci constructions combined with regions»; compute carefully with 11, 3.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.15.007

- (a) Accept accurate paraphrase of: Construct right and half-right angles. — *On-skill definition.*
- (b) Correct worked application of Construct 45° and 90° exam techniques using 8, 4 — *Use method from «Construct 45° and 90° exam techniques»; compute carefully with 8, 4.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.15.008

- (a) Accept accurate paraphrase of: ASA construction within exam tolerance. — *On-skill definition.*
- (b) Correct worked application of Construct triangle given ASA accurately using 20, 5 — *Use method from «Construct triangle given ASA accurately»; compute carefully with 20, 5.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.15.009

- (a) Accept accurate paraphrase of: Timed construction accuracy drill 1. — *On-skill definition.*
- (b) Correct worked application of Construction accuracy drill 1 using 7, 14 — *Use method from «Construction accuracy drill 1»; compute carefully with 7, 14.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.15.010

- (a) Accept accurate paraphrase of: Timed construction accuracy drill 2. — *On-skill definition.*
- (b) Correct worked application of Construction accuracy drill 2 using 10, 6 — *Use method from «Construction accuracy drill 2»; compute carefully with 10, 6.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.15.011

- (a) Accept accurate paraphrase of: Timed construction accuracy drill 3. — *On-skill definition.*
- (b) Correct worked application of Construction accuracy drill 3 using 6, 10 — *Use method from «Construction accuracy drill 3»; compute carefully with 6, 10.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.052

- (a) $(-1/2, \sqrt{3}/2) — (\cos \theta, \sin \theta).$

- (b) 60° — *Acute angle to x-axis.*
(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.053

- (a) $(-\sqrt{3}/2, 1/2)$ — $(\cos \theta, \sin \theta)$.
(b) 30° — *Acute angle to x-axis.*
(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.9.054

- (a) $(-1, 0)$ — $(\cos \theta, \sin \theta)$.
(b) 0° — *Acute angle to x-axis.*
(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.V.10.034

- (a) $\sqrt{3}/2$ — *Special triangles.*
(b) $\sqrt{2}/2$ — *Exact cosine.*
(c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.10.035

- (a) $1/2$ — *Special triangles.*
(b) $\sqrt{2}/2$ — *Exact cosine.*
(c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.6.032

- (a) $1/2$ — *Special triangles.*
(b) $\sqrt{2}/2$ — *Exact cosine.*
(c) $\sqrt{3}/2$ — *Identity $2\sin \cos = \sin 2\theta$, or multiply.*

T.V.7.023

- (a) 110 — *Pad to 3 digits; clockwise from north.*
(b) 290 — $\pm 180^\circ$.
(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*